

of  $f$ , denoted by  $c(f)$ , as any directed  $S$ -path from  $u$  to  $v$  that utilizes only forward arcs from  $A_B(T_\sigma, P_\sigma)$ .

► **Definition 13** (certification of an arc set). Consider an ordered partition  $P_\sigma = (V_1, \dots, V_\ell)$  of an ordered tournament  $T_\sigma$  and let  $F$  be a set of some backward arcs of  $T_\sigma$ . We say that *we can certify*  $F$  whenever it is possible to find a set  $\mathcal{F} = \{c(f) : f \in F\}$  of arc-disjoint certificates for the arcs in  $F$ .

► **Definition 14** (safe partition). Consider an ordered partition  $P_\sigma = (V_1, \dots, V_\ell)$  of an ordered tournament  $T_\sigma$ . Let  $B_E$  denote the set of all backward arcs in  $A_B(T_\sigma, P_\sigma)$ . For a vertex subset  $S \subseteq V(T)$ , we say  $P_\sigma$  is a  *$S$ -safe partition* if we can certify  $B_E$ .

Let  $D_\sigma = (V_\sigma, A)$  be an ordered directed graph. We use  $\text{Sfas}(D_\sigma, S)$  to denote the size of a minimum subset feedback arc set in the directed graph  $D_\sigma$ . Specifically, given a directed graph  $D_\sigma$ , and a vertex subset  $S \subseteq V_\sigma$ , it refers to the cardinality of the smallest set  $F \subseteq A$  of arcs whose removal (or reversal) renders  $D_\sigma$   $S$ -acyclic. Given a tournament, for an arc subset  $A' \subseteq A[T]$ , the directed graph  $T[A']$  is defined by the subgraph  $T$  containing the arc set  $A'$ . More specifically,  $T[A'] = (V(T), A')$ . For a ease of notation, when the subset is clear to the context, we use  $\text{Sfas}(D_\sigma)$  instead of  $\text{Sfas}(D_\sigma, S)$ .

► **Lemma 15.** *Consider an ordered partition  $P_\sigma = (V_1, \dots, V_\ell)$  of an ordered tournament  $T_\sigma$  and a terminal set  $S \subseteq V(T)$ . If  $P_\sigma$  is a  $S$ -safe partition then,*

$$\text{Sfas}(T_\sigma) = \text{Sfas}(T_\sigma[A_I(T_\sigma, P_\sigma)]) + \text{Sfas}(T_\sigma[A_B(T_\sigma, P_\sigma)])$$

*Moreover, there exists a minimum sized  $S$ -feedback arc set of  $T_\sigma$  containing  $B_E$ , where  $B_E$  denote the set of all backward arcs in  $A_B(T_\sigma, P_\sigma)$ .*

**Proof.** Given any bipartition of the arc set  $A$  into  $A_1$  and  $A_2$ , we have

$$\text{Sfas}(T_\sigma) \geq \text{Sfas}(T_\sigma[A_1]) + \text{Sfas}(T_\sigma[A_2]).$$

Specifically, for the partition of  $A$  into  $A_I(T_\sigma, P_\sigma)$  and  $A_B(T_\sigma, P_\sigma)$ , it follows that

$$\text{Sfas}(T_\sigma) \geq \text{Sfas}(T_\sigma[A_I(T_\sigma, P_\sigma)]) + \text{Sfas}(T_\sigma[A_B(T_\sigma, P_\sigma)]).$$

Next, we need to show that

$$\text{Sfas}(T_\sigma) \leq \text{Sfas}(T_\sigma[A_I(T_\sigma, P_\sigma)]) + \text{Sfas}(T_\sigma[A_B(T_\sigma, P_\sigma)]).$$

This assertion holds because, after reversing all arcs in  $B_E$ , each remaining directed  $S$ -cycle is contained within  $T_\sigma[V_i]$  for some  $i \in [\ell]$ . In other words, once all arcs in  $B_E$  are reversed, every  $S$ -cycle is entirely contained within  $T_\sigma[A_I(T_\sigma, P_\sigma)]$ . Observe that as  $P_\sigma$  is  $S$ -safe, the set of all backward arcs of  $A_B(T_\sigma, P_\sigma)$ , i.e.,  $B_E$  can be certified using only arcs from  $A_B(T_\sigma, P_\sigma)$ . This concludes the proof of the first part of the lemma. Essentially, we have shown the existence of a minimum-sized  $S$ -feedback arc set for  $T_\sigma$  that includes  $B_E$ . This completes the proof of the lemma. ◀

**Construction of a  $S$ -safe partition.** Recall that an  $S$ -triangle is a directed cycle of length three that includes at least one vertex from  $S$ . It is clear that the number of arc-disjoint  $S$ -triangles provides a lower bound for the size of the smallest  $S$ -feedback arc set in a tournament. We illustrate how this set can be utilized to identify a safe partition in polynomial time.